

Histograms and Cumulative Frequency

GCSE Maths · Worked Solutions

20 questions · 62 marks total

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Section A · Foundation

Questions 1–3 are the most straightforward; they get harder as you go on. Question 10 is the most challenging.

1 [1 mark]

In a histogram where every class has the same width, what does the height of each bar represent?

- 1 With equal class widths, the frequency density is just the frequency divided by the same number every time.
- 2 So the heights are simply proportional to the frequencies.

Answer: the frequency of that class

2 [2 marks]

Complete the cumulative frequency column.

Class	Frequency	Cumulative frequency
$0 < x \leq 10$	5	
$10 < x \leq 20$	9	
$20 < x \leq 30$	12	
$30 < x \leq 40$	4	

- 1 Cumulative frequency is a running total.
- 2 5, then $5 + 9 = 14$, then $14 + 12 = 26$, then $26 + 4 = 30$

Answer: 5, 14, 26, 30

3 [2 marks]

Using the completed table above,

- (a) how many values are 30 or less?
- (b) how many values are more than 20?

- 1 **(a)** Read the cumulative frequency for $x \leq 30$: 26
- 2 **(b)** The total is 30, and 14 values are 20 or less.
- 3 $30 - 14 = 16$

Answer: (a) 26 (b) 16

4 [2 marks]

Using the same table (total frequency 30), write down the class interval that contains the median.

- 1 The median is between the 15th and 16th values.
- 2 The cumulative frequencies are 5, 14, 26, 30, so the 15th and 16th values fall in the third class.

Answer: $20 < x \leq 30$

5

[3 marks]

A frequency polygon is to be drawn for this table.

Class	0–10	10–20	20–30	30–40
Frequency	8	14	11	5

Write down the four points that should be plotted.

- 1 A frequency polygon is plotted at the **midpoint** of each class.
- 2 Midpoints: 5, 15, 25 and 35
- 3 Pair each midpoint with its frequency.

Answer: (5, 8), (15, 14), (25, 11) and (35, 5)

6

[3 marks]

Work out the frequency density for each class.

Class width	10	10	10	10
Frequency	12	25	18	5

- 1 Frequency density = $\frac{\text{frequency}}{\text{class width}}$
- 2 $12 \div 10 = 1.2$ and $25 \div 10 = 2.5$
- 3 $18 \div 10 = 1.8$ and $5 \div 10 = 0.5$

Answer: 1.2, 2.5, 1.8 and 0.5

7

[3 marks]

Class A is $0 < x \leq 10$ with frequency 20. Class B is $10 < x \leq 30$ with frequency 30.

- (a) Work out the frequency density of each class.
- (b) Which class has the values packed more closely together? Explain.

- 1 (a) Class A: $20 \div 10 = 2$ · Class B: $30 \div 20 = 1.5$
- 2 (b) Class B has more values, but they are spread over a much wider range.
- 3 Class A has the higher frequency density, so its values are packed more closely.

Answer: (a) A: 2, B: 1.5 (b) class A, because it has the greater frequency density

8

[3 marks]

The table shows cumulative frequencies for the heights, h cm, of 50 plants.

Height	$h \leq 160$	$h \leq 170$	$h \leq 180$	$h \leq 190$
Cumulative frequency	12	30	46	50

Estimate how many plants are 175 cm or less.

- 1 175 is halfway between 170 and 180.
- 2 The cumulative frequency rises from 30 to 46 across that class, an increase of 16.
- 3 Assuming the values are spread evenly, half of that increase has happened by 175: $\frac{1}{2} \times 16 = 8$
- 4 Estimate = $30 + 8 = 38$ plants

Answer: about 38 plants

9

[4 marks]

The table shows cumulative frequencies for 60 values.

Value (x)	$x \leq 10$	$x \leq 20$	$x \leq 30$	$x \leq 40$	$x \leq 50$
Cumulative frequency	10	25	45	55	60

- (a) Estimate the median.
 (b) Estimate the lower quartile.
 (c) Estimate the interquartile range.

1 (a) The median is the 30th value (half of 60). It lies between 20 and 30, where the cumulative frequency goes from 25 to 45.

2 It is the $30 - 25 = 5$ th of those 20 values: $20 + \frac{5}{20} \times 10 = 22.5$

3 (b) The lower quartile is the 15th value, between 10 and 20 (cumulative 10 to 25).

4 $10 + \frac{5}{15} \times 10 = 13.3$ (1 d.p.)

5 (c) The upper quartile is the 45th value, which the table shows is exactly at $x = 30$.

6 $IQR = 30 - 13.3 = 16.7$

Answer: (a) about 22.5 (b) about 13.3 (c) about 16.7

10

[4 marks]

A histogram has these bars.

Class	0–5	5–10	10–20	20–40
Frequency density	4	6	3	1

- (a) Work out the frequency of each class and the total.
 (b) Estimate how many values are greater than 15.
- 1 (a) Frequency = frequency density $\times$ class width.
 2 $4 \times 5 = 20$, $6 \times 5 = 30$, $3 \times 10 = 30$, $1 \times 20 = 20$
 3 Total = $20 + 30 + 30 + 20 = 100$
 4 (b) From 15 to 20 is half of the 10–20 class: $3 \times 5 = 15$ values.
 5 The whole 20–40 class adds 20 more, so the estimate is $15 + 20 = 35$

Answer: (a) 20, 30, 30, 20 (total 100) (b) about 35

Section B · Higher

Higher-tier style. Again, questions 1–3 are the gentlest and question 10 is the hardest.

1

[2 marks]

A class is $20 < x \leq 50$ and its frequency is 45.

Work out the frequency density.

1 Class width = $50 - 20 = 30$

2 Frequency density = $\frac{45}{30} = 1.5$

Answer: 1.5

2

[2 marks]

A histogram bar has frequency density 2.4 and covers a class of width 5.

Work out the frequency of that class.

1 Frequency = frequency density $\times$ class width

2 $= 2.4 \times 5 = 12$

Answer: 12

3

[3 marks]

Complete the frequency density column.

Class width	6	11	13	24
Frequency	18	33	26	12

1 Divide each frequency by its class width.

2 $18 \div 6 = 3$ and $33 \div 11 = 3$

3 $26 \div 13 = 2$ and $12 \div 24 = 0.5$

Answer: 3, 3, 2 and 0.5

4

[3 marks]

A class $20 < x \leq 40$ has frequency 36.

Estimate how many values lie between 25 and 40.

1 The class is 20 wide and the part we want, from 25 to 40, is 15 wide.

2 Assume the values are spread evenly across the class.

3 $\frac{15}{20} \times 36 = 0.75 \times 36$

4 $= 27$ values

Answer: about 27

5

[4 marks]

A histogram has these bars.

Class	0–10	10–20	20–50	50–80
Frequency density	1.5	2.8	1.0	0.4

(a) Work out the total frequency.

(b) Estimate the median.

1 (a) Frequencies: $1.5 \times 10 = 15$, $2.8 \times 10 = 28$, $1.0 \times 30 = 30$, $0.4 \times 30 = 12$

2 Total $= 15 + 28 + 30 + 12 = 85$

3 (b) The median is the 42.5th value (half of 85).

4 Running totals: 15 up to 10, then $15 + 28 = 43$ up to 20, so the median is in 10–20.

5 It is the $42.5 - 15 = 27.5$ th of the 28 values in that class.

6 Median $\approx 10 + \frac{27.5}{28} \times 10 = 10 + 9.82 = 19.8$

Answer: (a) 85 (b) about 19.8

6

[4 marks]

The table shows cumulative frequencies for 80 values.

Value (x)	$x \leq 20$	$x \leq 40$	$x \leq 60$	$x \leq 80$	$x \leq 100$
Cumulative frequency	8	26	54	72	80

(a) Estimate the median.

(b) Estimate the quartiles.

(c) Work out the interquartile range.

1 (a) The median is the 40th value, lying between 40 and 60 (cumulative 26 to 54).

$$2 \quad 40 + \frac{14}{28} \times 20 = 40 + 10 = 50$$

3 (b) Lower quartile: the 20th value, between 20 and 40 (cumulative 8 to 26).

$$4 \quad 20 + \frac{12}{18} \times 20 = 20 + 13.3 = 33.3$$

5 Upper quartile: the 60th value, between 60 and 80 (cumulative 54 to 72).

$$6 \quad 60 + \frac{6}{18} \times 20 = 60 + 6.7 = 66.7$$

7 (c) IQR = $66.7 - 33.3 = 33.4$ (about 33)

Answer: (a) 50 (b) LQ about 33.3, UQ about 66.7 (c) about 33.4

7

[4 marks]

Using the same cumulative frequency table (80 values),

(a) estimate how many values are greater than 70.

(b) express that as a percentage of the total.

1 (a) First estimate the cumulative frequency at 70, which is halfway between 60 and 80.

2 The cumulative frequency rises from 54 to 72 there, an increase of 18.

$$3 \quad \text{Halfway gives } 54 + \frac{1}{2} \times 18 = 54 + 9 = 63 \text{ values at 70 or below.}$$

$$4 \quad \text{Values above 70} = 80 - 63 = 17$$

$$5 \quad (b) \quad \frac{17}{80} \times 100 = 21.25\%$$

Answer: (a) about 17 (b) about 21.25%

8

[4 marks]

Two groups of test marks have these summary values.

	Median	Lower quartile	Upper quartile
Group P	54	41	68
Group Q	60	54	67

(a) Work out the interquartile range for each group.

(b) Compare the two groups, giving two clear comparisons.

1 (a) Group P: $68 - 41 = 27$ · Group Q: $67 - 54 = 13$

2 (b) Averages: Group Q has the higher median (60 against 54), so on average Group Q scored better.

3 Spread: Group Q has the smaller interquartile range (13 against 27), so its marks are more consistent — the middle half of Q's marks are packed into a much narrower band.

Answer: (a) P: 27, Q: 13 (b) Q has the higher median (60 v 54) and the smaller IQR (13 v 27), so it scored higher and more consistently

9

[4 marks]

A histogram has three bars. The first, for $0 < x \leq 20$, has frequency density 1.2. The second, for $20 < x \leq 30$, has frequency 30. The third covers $30 < x \leq 60$.

The total frequency is 90. Work out the frequency density of the third bar.

1 First bar frequency = $1.2 \times 20 = 24$

2 Second bar frequency = 30 (given).

3 Third bar frequency = $90 - 24 - 30 = 36$

4 The third class width is $60 - 30 = 30$

5 Frequency density = $\frac{36}{30} = 1.2$

Answer: 1.2

A histogram of the times, t minutes, taken by 80 people has these bars.

Time (t min)	$0 < t \leq 10$	$10 < t \leq 20$	$20 < t \leq 40$	$40 < t \leq 80$
Frequency density	1.2	2.8	1.4	0.3

(a) Work out the frequency of each class and show the total is 80.

(b) Estimate the median.

(c) Estimate how many people took more than 30 minutes.

(d) Estimate the interquartile range.

1 (a) $1.2 \times 10 = 12$, $2.8 \times 10 = 28$, $1.4 \times 20 = 28$, $0.3 \times 40 = 12$

2 Total = $12 + 28 + 28 + 12 = 80$ ✓

3 (b) The median is the 40th value. Running totals are 12, 40, 68, 80.

4 The 40th value is exactly the last of the second class, so the median is about 20 minutes.

5 (c) From 30 to 40 is half of the third class: $\frac{1}{2} \times 28 = 14$

6 Plus the whole final class of 12, giving $14 + 12 = 26$ people.

7 (d) Lower quartile: the 20th value, in the second class (12 to 40).

8 $10 + \frac{8}{28} \times 10 = 12.9$ (1 d.p.)

9 Upper quartile: the 60th value, in the third class (40 to 68): $20 + \frac{20}{28} \times 20 = 34.3$

10 IQR $\approx 34.3 - 12.9 = 21.4$ minutes

Answer: (a) 12, 28, 28, 12 (total 80) (b) about 20 minutes (c) about 26 people (d) about 21.4 minutes